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PID, properly

Week 3 · CADE30008 Flight Dynamics & Advanced Control

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Today

Two staff, two aircraft, and a method that stops working.

- **An aircraft you already know:** measured, not simulated
- **An aircraft you don't**, with four poles: tune it yourself then we'll review
- **Challenges** of controlling more complex models
- **The Quanser laboratory**: top tips on what a good session looks like

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What you already know

Recap and a familiar case · 20 min

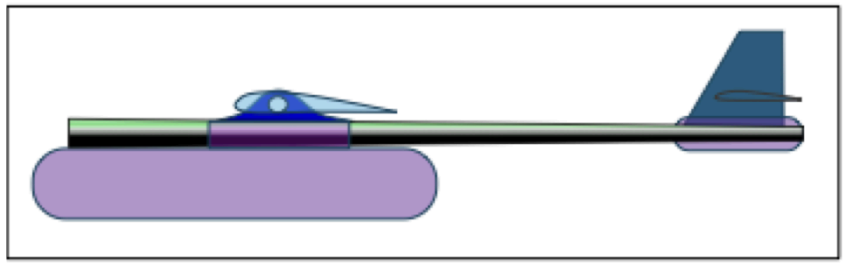
PID, from last year

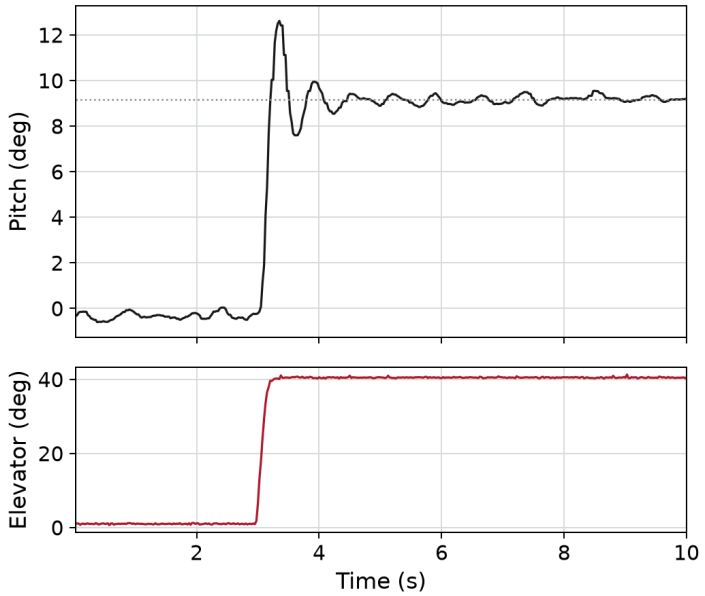
Term	What it does	What it costs
P	Lifts the magnitude, moves crossover up	Less damping as gain rises
I	Removes steady-state error, adds a type	Phase, right where you need it
D	Buys phase lead around crossover	Noise, and a kick on a step

You did all of this last year, in the signals, systems and control theme: the three terms, steady-state error for P, PD and PID, Ziegler-Nichols, and matching the characteristic equation.

A familiar aircraft: pitch controller design

YOU DO





A small UAV wind-tunnel model. `pitch_step_response.mat`, in the Drive folder.

1. **Fit a transfer function** to the pitch response
2. **Design and tune a PI or PID controller** against your fit, to track a reference
3. Say how you would know whether it is any good

In pairs, fifteen minutes. Done when you have gains you would defend.

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An aircraft you have not met

Tune it yourself · 20 min

Motivations

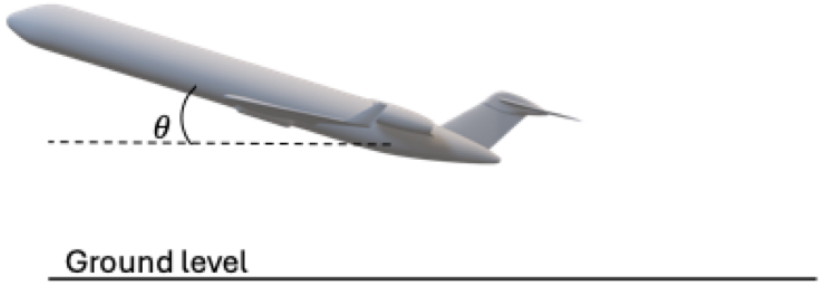
- Provide an example of **when we need to go beyond PID**
- Illustrate what **'full fidelity' pitch dynamics** look like in a fixed-wing aircraft
- Show that **we're smarter than MATLAB's PID tuner**

YOU DO

Aircraft pitch dynamics

Pitch dynamics has 4 poles and (usually) two zeros.

Develop a PID controller for this system. You may use whatever tools you have at your disposal.



$$\frac{\theta(s)}{\delta_{\mathrm{e}}(s)} = \frac{(s+0.5877)\,(s+0.02662)}{(s^2+1.127s+1.862)(s^2+0.01579s+0.01475)}\left(\frac{\mathrm{deg}}{\mathrm{deg}}\right)$$

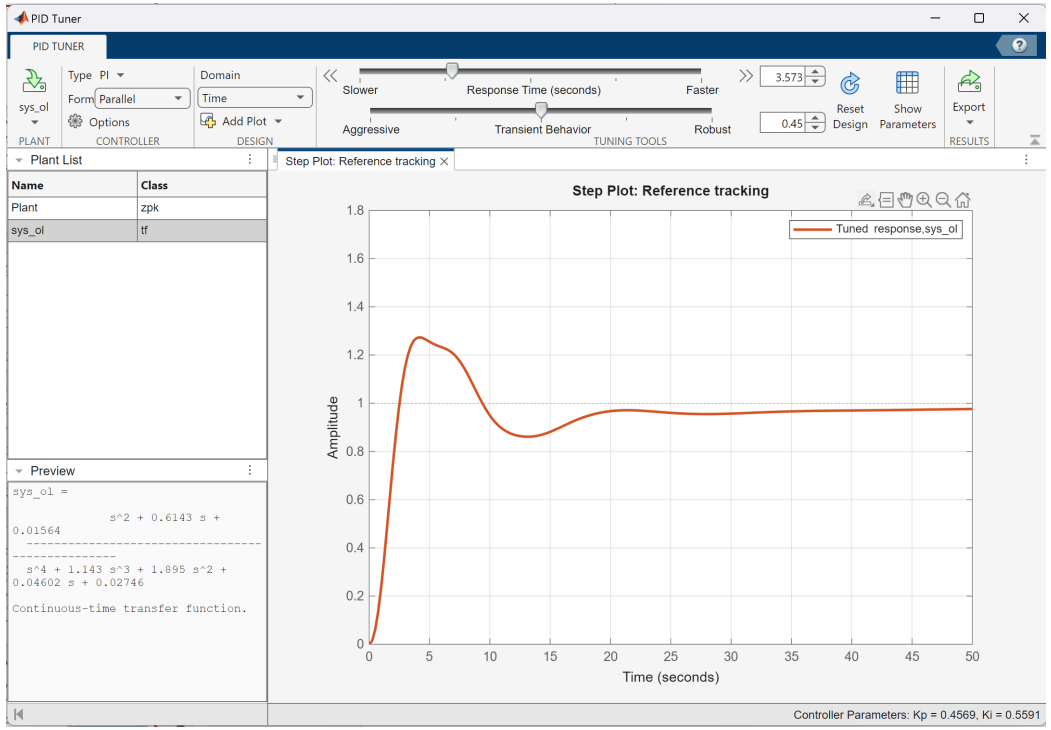
Minus sign (negative elevator generates pitch up moment) omitted here for simplicity.

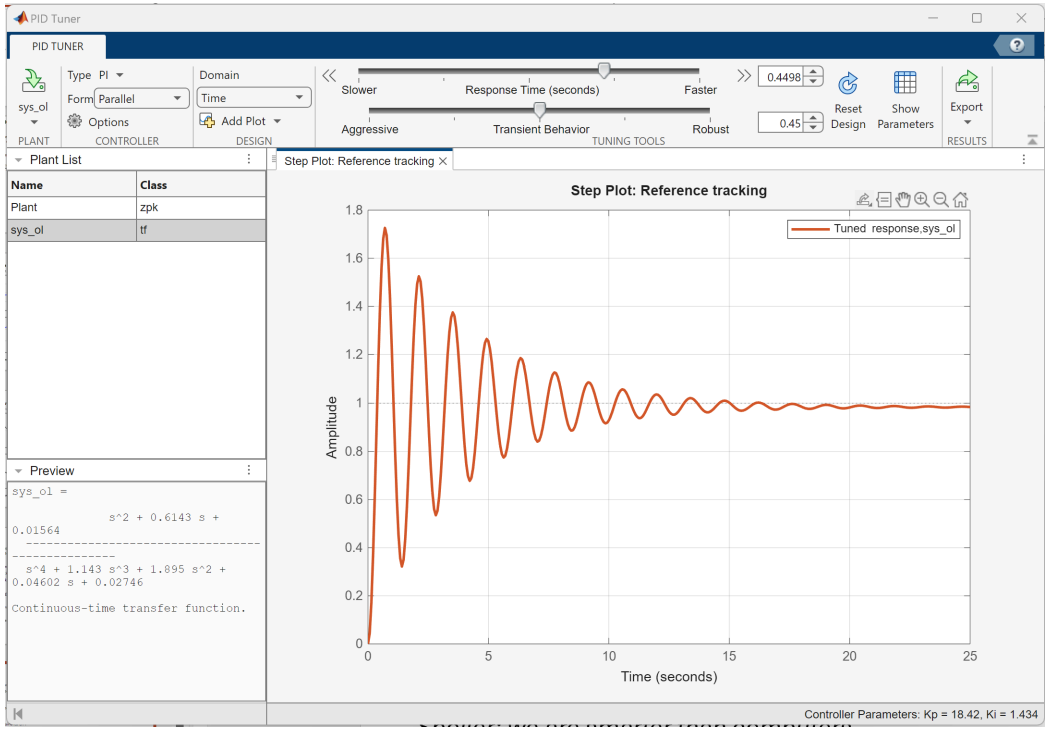
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Break

Ten minutes

MATLAB's auto PI tuning





More stable: 18 s to 5%, 57 s to 2%. **Too slow.**

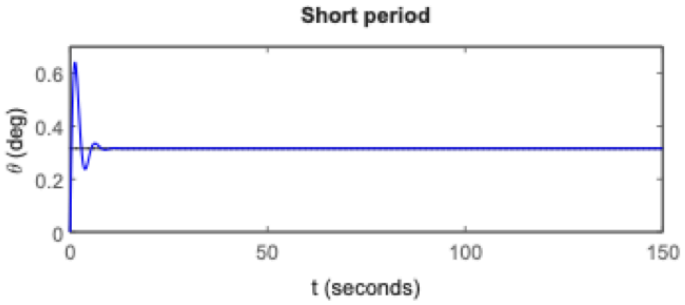
Faster response: but not enough damping

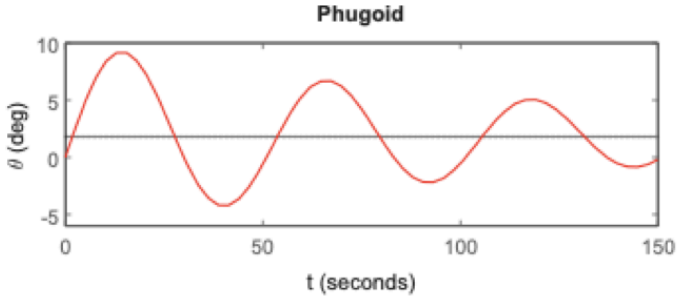
Why?

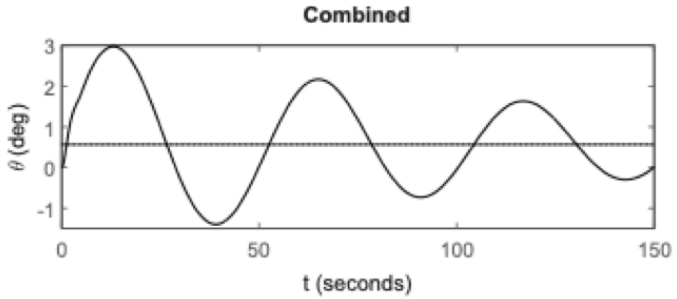
Open loop dynamics

$$\frac{\theta(s)}{\delta_e(s)} = \underbrace{\frac{s + 0.5877}{s^2 + 1.127s + 1.862}}_{\text{short period}} \times \underbrace{\frac{s + 0.02662}{s^2 + 0.01579s + 0.01475}}_{\text{phugoid}}$$

Fourth order dynamics. Two modes spanning two different time frames.







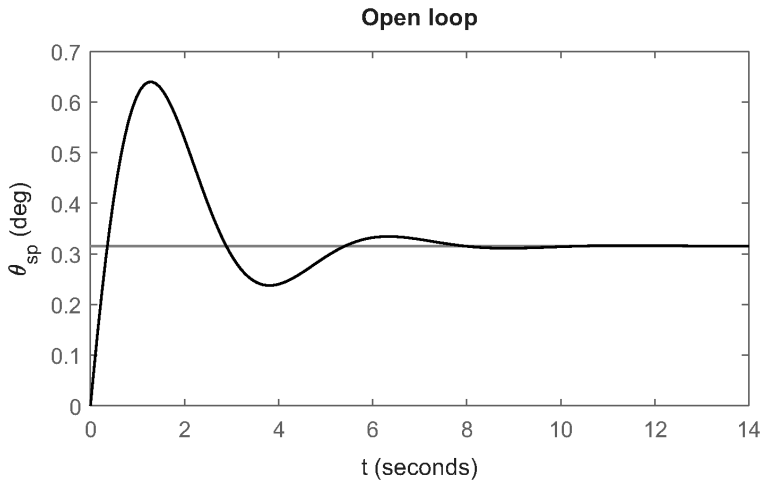
Short period dynamics

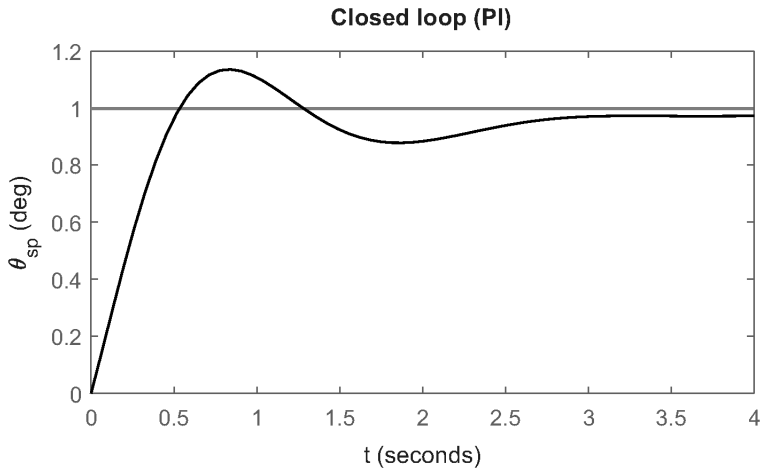
If we ignore the phugoid mode, the dynamics reduces to a simple second-order system.

$$\frac{\theta(s)}{\delta_{\mathrm{e}}(s)} = \underbrace{\frac{s + 0.5877}{s^2 + 1.127s + 1.862}}_{\text{short period}}$$

PI works well here. MATLAB's control tuner app gave me a PI controller with gains K_{p} = 2.2, K_{i} = 8.3.

$$C(s) = 2.2 + \frac{8.3}{s}$$





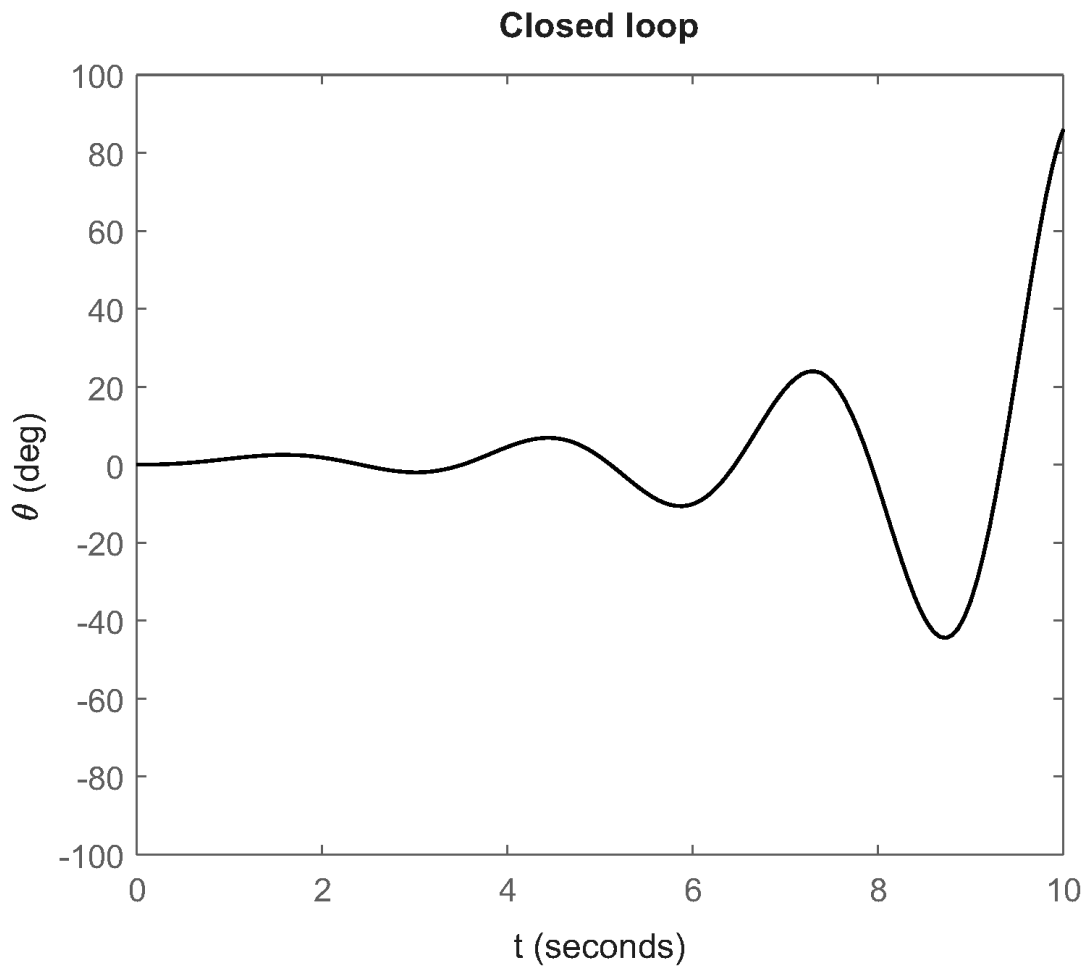
Combined closed-loop dynamics

Now use the same controller on the full transfer function with both fast and slow dynamics

Open-loop transfer function

$$C(s)\frac{\theta(s)}{\delta_e(s)} = \left(2.2 + \frac{8.3}{s}\right)\left(\frac{s+0.5877}{s^2+1.127s+1.862} \times \frac{s+0.02662}{s^2+0.01579s+0.01475}\right)$$

What's going on?

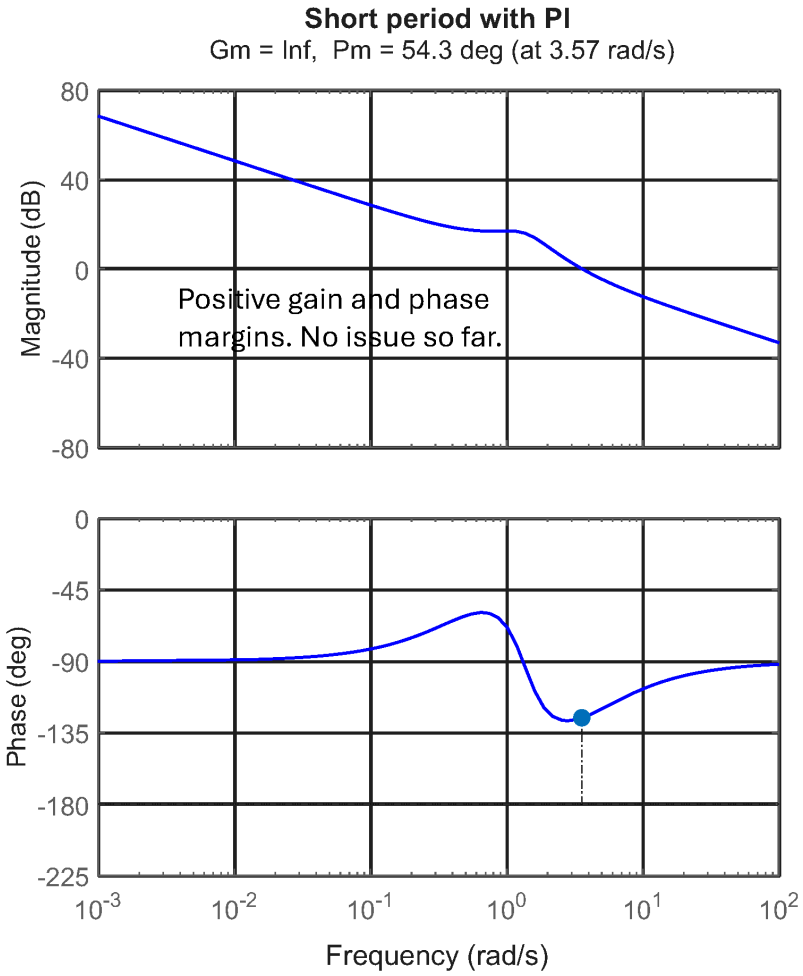


Why it failed

Frequency response: PI controller + short period only

$$C(s)\,\frac{\theta(s)}{\delta_{\mathrm{e}}(s)} = \left(2.2 + \frac{8.3}{s}\right)\left(\frac{s + 0.5877}{s^2 + 1.127s + 1.862}\right)$$

Positive gain and phase margins. No issue so far.

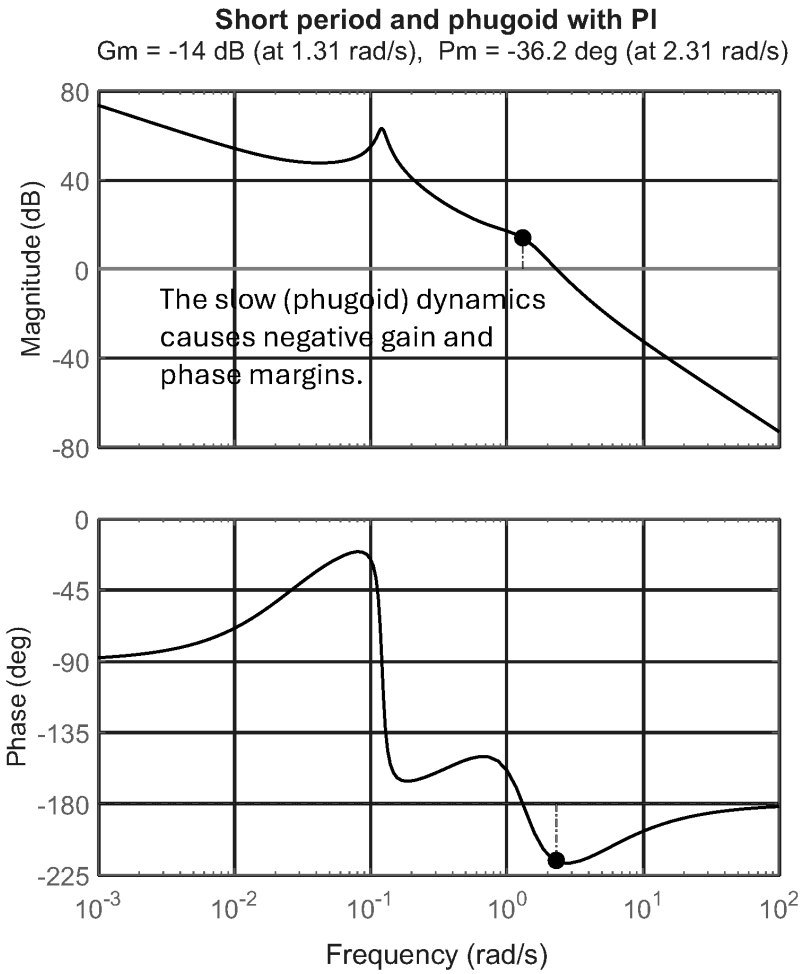


Why it failed

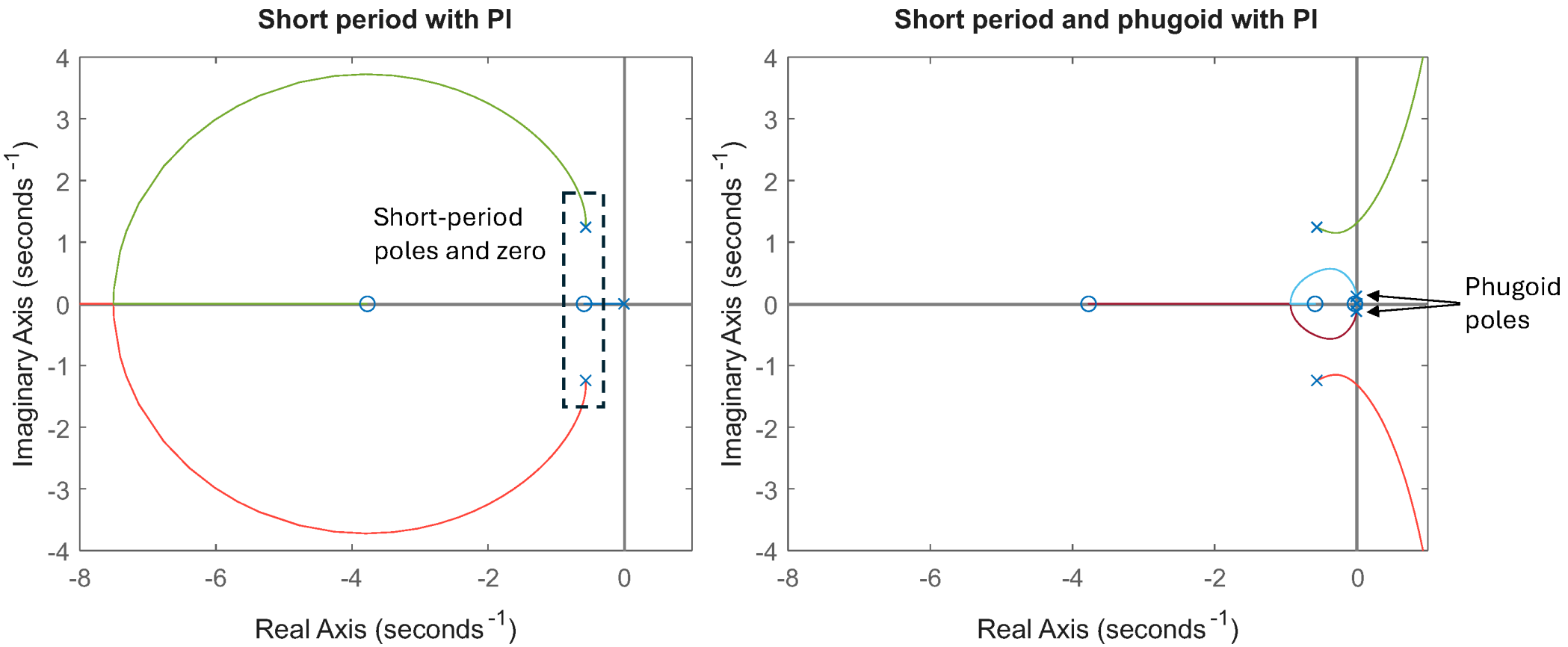
Frequency response: PI controller + full dynamics

$$C(s)\frac{\theta(s)}{\delta_e(s)} = \left(2.2 + \frac{8.3}{s}\right)\left(\frac{s + 0.5877}{s^2 + 1.127s + 1.862} \times \frac{s + 0.02662}{s^2 + 0.01579s + 0.01475}\right)$$

The slow (phugoid) dynamics causes negative gain and phase margins.



Why it failed



We have too many poles and zeros to manipulate.

Summary

For higher-order system, simply using PID can lead to unsatisfactory performance.

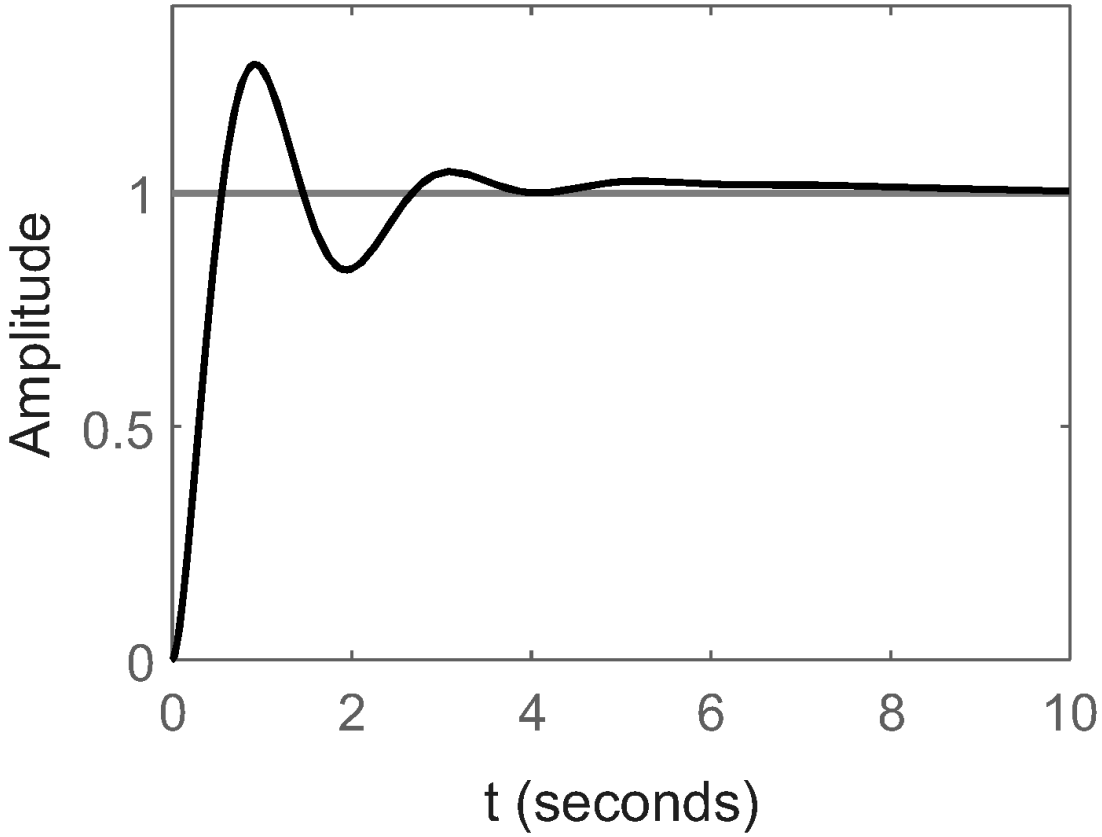
More serious if the system has dynamics spanning different time frames.

Solutions?

Loop shaping (transfer function based).

Multi-variable feedback (state space).

Teaser: with just one extra transfer function on top of the existing PI controller, the response on the right can be achieved.



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The Quanser laboratory

Demo and tips · 20 min

Today, in four lines

- **PID is a method, not an answer.** It was revision, and it still failed
- **A controller is designed against a model**, and inherits everything the model got wrong
- **Two time scales in one plant** are what beat it today
- **More PID tuning detail is in the live script**, `L03_tune_and_saturate.m`: tune it, then saturate the elevator

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It meets every requirement

How close to the edge is it?

See you next session

(after consolidation week - next week: guest lecture)